

Thai Poetry Rhyme Verification using Hybrid Rules Based Approach for Education*

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1st Warit Yuvaniyama

*School of Information, Computer, and Communication Technology
Sirindhorn International Institute of Technology, Thammasat University
Pathum Thani 12120, Thailand
<https://orcid.org/0009-0007-1630-4286>*

2nd Athit Phinyachok

*School of Information, Computer, and Communication Technology
Sirindhorn International Institute of Technology, Thammasat University
Pathum Thani 12120, Thailand
<https://orcid.org/0009-0000-3783-8104>*

3rd Gunn Sanguankittipun

*School of Information, Computer, and Communication Technology
Sirindhorn International Institute of Technology, Thammasat University
Pathum Thani 12120, Thailand
<https://orcid.org/0009-0005-8812-4992>*

4th Leo Colombo

*School of Information, Computer, and Communication Technology
Sirindhorn International Institute of Technology, Thammasat University
Pathum Thani 12120, Thailand
<https://orcid.org/0009-0004-9133-5398>*

5th Prachya Boonkwan*

*School of Information, Computer, and Communication Technology
Sirindhorn International Institute of Technology, Thammasat University
Pathum Thani 12120, Thailand
<https://orcid.org/0000-0003-1405-1071>**

Abstract—

Index Terms—Thai NLP, Poetry Generation, Rule-based Verification, Natural Language Processing, Reinforcement Learning, Educational Technology

I. INTRODUCTION

- **Context:** Introduce Thai Klon-Paed () and its strict structural rules (Sumpus, Mattrra, Sara).
- **The Core Problem:** Generative AI and LLMs are probabilistic, making them inherently bad at strict deterministic constraints like syllables and rhymes.
- **The Dual Need:**
 - *Research:* To train an LLM to write poetry (e.g., via RLHF/RLAIF), we need a flawless, programmatic reward environment to score rhymes.

- *Education:* Teachers need zero-hallucination, interactive grading tools for students.

- **Limitations of Prior Art:** Previous checkers (PyThaiNLP 5.0.1) and G2P models (Kongfha) fail on edge cases, drop stanzas, or are computationally too slow for RL loops.
- **Contributions:**
 - Improved core verification algorithms (`check_sara`, `check_mattrra`, `is_sumpus`).
 - Creation of a 36,475-stanza gold-standard evaluation dataset.
 - Integration into PyThaiNLP 5.3.5 and an open-source Hugging Face educational application.

II. BACKGROUND AND RELATED WORK

A. Linguistic Rules of Klon-Paed

- Briefly define the requirements for a valid rhyme in Thai: matching vowel sound (`check_sara`) and matching final consonant class (`check_marttra`).
- Define inter-stanza (`rX`) and intra-stanza (`r1`, `r2`, `r3`) connections.

B. Existing Verification Systems

- **Model A (Old PyThaiNLP 5.0.1):** Dictionary-based, lacks the `r3` rule, fails on complex morphology.
- **Model C (Kongfha / tltk G2P):** Romanization-based. Extremely slow (single-threaded) and collapses short/long vowels, leading to false positives.

III. METHODOLOGY: HYBRID VERIFICATION ARCHITECTURE

A. Core Rule-Based Improvements (Model B)

- Explain the mechanics of Model B (now PyThaiNLP 5.3.5).
- **Phonetic Normalization:** Handling transformed/reduced vowels (`,` `.`).
- **Edge Case Handling:**
 - `handle_karun_sound_silence`: Stripping silenced characters (e.g., `''' → ''`).
 - `_is_true_final`: Differentiating genuine finals (`-`, `-`) from clusters or digraphs.

B. Dictionary-Enhanced Verification (Model D)

- Explain the Klonpad variant. While Model B acts as a general-purpose oracle, it has "blind spots" (e.g., the silent `''` in `'''`).
- Introduce the G2P Override Dictionary (built on Phra Aphai Mani) to fix these specific phonetic misreadings.
- **Ablation Study of Fallbacks:** Briefly discuss why `D_ssg` (SSG fallback) outperformed `D_w2p` (W2P fallback). W2P introduces hallucinations on short words, hurting overall accuracy.

IV. DATASET CONSTRUCTION AND AUGMENTATION

A. Gold-Standard Corpus Compilation

- Data sourced from vajirayana.org (Khun Chang Khun Phaen, Phra Aphai Mani, etc.).
- Cleaning pipeline resulting in 191 files, 36,475 stanzas, and 145,709 total rhyme checks.
- Highlight the data-completeness fix (retaining ≥ 10 syllable waks) that restored inter-stanza (`rX`) chains.

B. Targeted Data Augmentation

- Explain the need for precision and recall probes.
- **Negative Sampling (Precision Probes):** 10,000 instances. Discuss operators like `C3_short_long` (vowel length swaps) and `C9_old_accept` (traps for Model A).
- **Positive Sampling (Recall Probes):** 1,200 instances. Discuss "Oracle-blind" positives (e.g., `,` `.`) intentionally

designed to deduct points from Model B where it fails linguistically.

V. EVALUATION AND RESULTS

A. Experimental Setup

- Define the evaluated models (A, B, C, `D_w2p`, `D_ssg`) and metrics (Precision, Recall, F1, Coverage).
- Mention computational mitigations for Model C (persistent worker pool, G2P caching).

B. Gold Recall Analysis

- Present the Stanza-level gold recall table.
- Highlight that `D_ssg` achieves the highest recall (88.5%) followed by B (87.5%), vastly outperforming A (73.4%).

C. Augmentation and Error Analysis

- Present the Augmentation-only overall table.
- **Precision vs. Recall Trade-off:** Explain that Model B acts as a perfect oracle (0 FPs on negatives) but misses oracle-blind positives. Model C catches rhymes but has 3,129 FPs (mostly on `C3_short_long`).
- **The Override Advantage:** Show how Model `D_w2p` recovers 95.7% of the oracle-blind spots (e.g., reading `→` `)` that Model B misses.

VI. APPLICATIONS AND FUTURE WORK

A. Educational Impact

- Discuss the interactive Hugging Face web interface.
- Explain how deterministic, explainable feedback aids teachers and students in learning Klon 4/8 structures without LLM hallucinations.

B. Foundation for Generative AI

- Discuss how this verifier is currently implemented as the Reward Model environment for Reinforcement Learning (RL) to train a small LLM to write Klon-Paed.

VII. CONCLUSION

- Summarize: The hybrid approach effectively marries the reliability of rule-based logic with the edge-case coverage of G2P overrides.
- Final impact statement: The successful integration of Model B into PyThaiNLP 5.3.5 sets a new standard for Thai computational linguistics, paving the way for both educational tools and advanced LLM pipelines.

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- [3] [Add other relevant Thai NLP, G2P, or LLM RLHF papers here]